JAVA using BlueJ

Unit 1

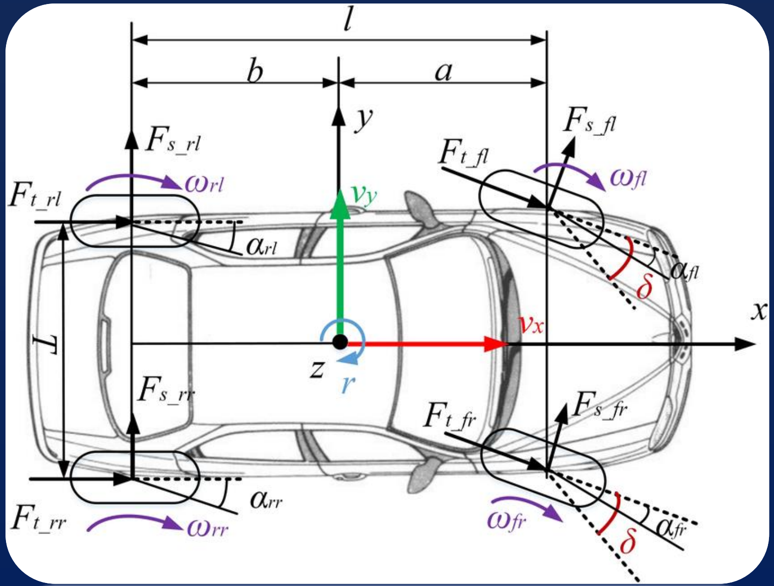
Elementary Concept of OOPS

Object Oriented Programming

- Object-oriented Programming System(OOPs) is a programming technique that is based on the **Class and Object** Model.
- In OPPS first we create layout(blue print) then implement (create) it.
- The primary goal of object-oriented programming is to increase the **flexibility** and **maintainability** of programs.
- OOPs is enable to solve the **real life problem** through program.

Class

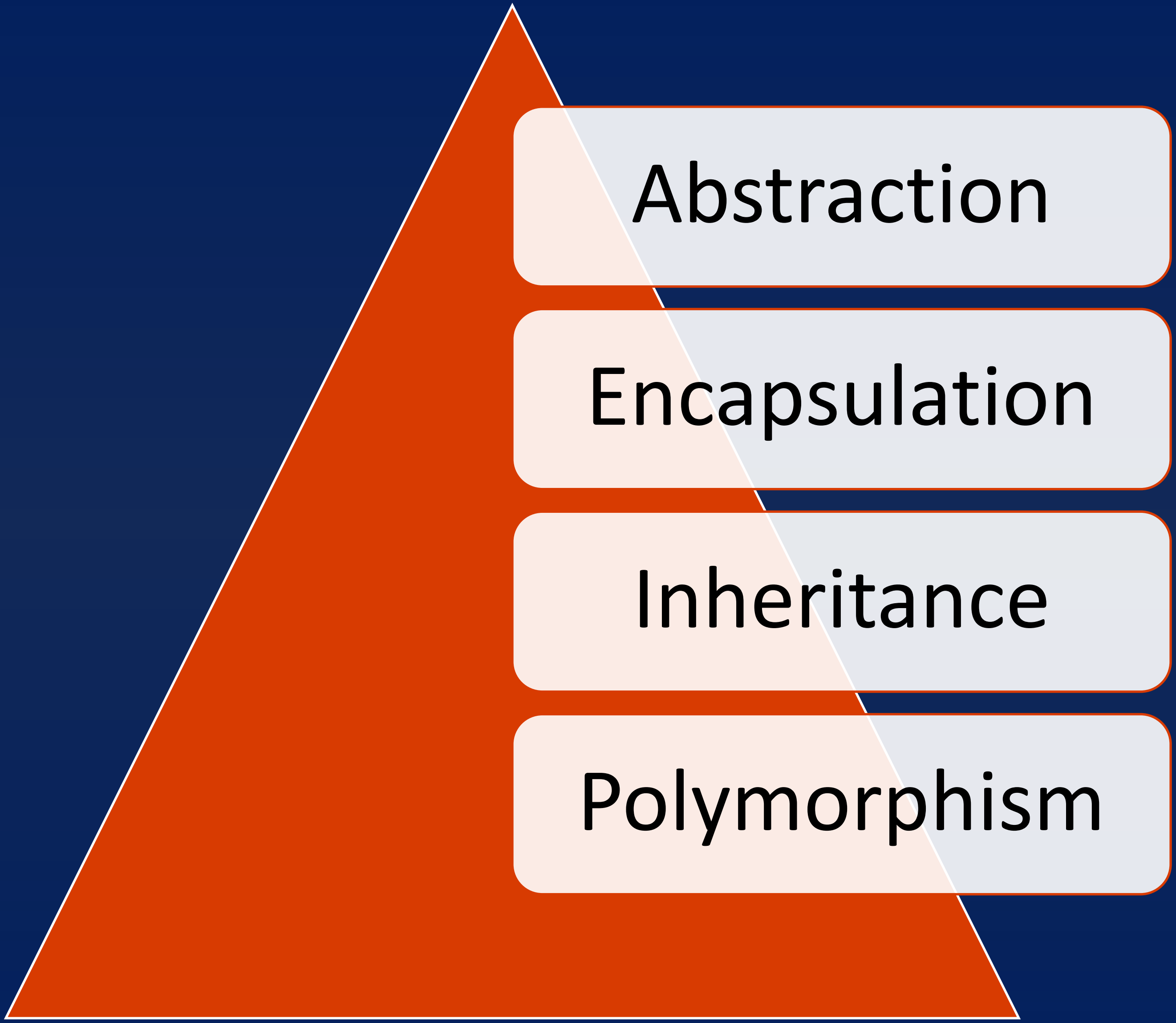
- A class can be defined as a template/blueprint that describes the behavior/state that the object of its type support.
- It is also known as ***User defined data Type***.
- It is only a **logical component** and not the physical entity.



Object

- An object can be defined as an **instance of a class**, and there can be multiple instances of a class in a program.
- Objects have states and behaviors.
- It has real life (physical) existence.

Principal of OOP



Abstraction

- Abstraction is the act of representing the essential components and hiding the background details.
- It is a process where you show only “relevant” data and “hide” unnecessary details of an object from the user.
- This can be done by access specifiers.

Encapsulation

The process of wrapping up data and associated functions together into a single unit (class) is known as encapsulation.

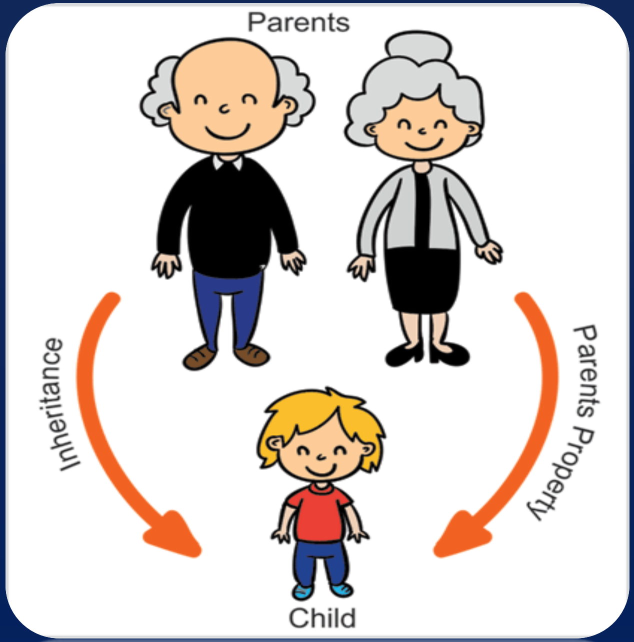
The basic functions of encapsulation are:

- Binding the data with the code that manipulates it.
- Keeping the data and the code safe from external interface.

- It can also known as **Data Binding**.

Inheritance

- Inheritance is a way to reuse once written code again and again.
- The class which is inherited is called base calls & the class which inherits is called derived class. So when, a derived class inherits a base class, the derived class can use all the functions which are defined in base class, hence making code reusable.



Polymorphism

- Polymorphism is the concept of using the same function name multiple
- Polymorphism is a Greek word which means "several forms“ or “Many Forms”.

Benefits of OOPS

OOP is a programming approach ,which offers a number of benefits

- Software complexity can easily be managed as it models real world object
- You can easily modify and maintain complexity of program.
- Data hiding can be used to create secure program.
- Classes can be reused for creating other classes through inheritance
- It is easy to partition the work in project based on object.

Question Answer

- 1. A set of instructions given to a computer to do a particular task.
  - a. Program
  - b. High level Language
  - c. Object
  - d. None of these
- 2. An object is represented by two attributes, out of which one is characteristics and the other one is ____________.
  - a. Behaviour
  - b. Situation
  - c. Abstraction
  - d. Encapsulation
- 3. Name the programming technique that implements programs as an organized collection of interactive objects.
  - a. Procedure Oriented Programming
  - b. Modular Programming
  - c. Object Oriented Programming
  - d. None of these
- 4. Name the programming technique that specifies a series of well-structured steps and procedures within its programming context to compose a program.
  - a. Procedure Oriented Programming
  - b. Modular Programming
  - c. Object Oriented Programming
  - d. None of these

Question Answer

- 5. Name the characteristics of Object Oriented Programming that hides the complexity and provides a simple interface.
  - a. Encapsulation b. Polymorphism
  - c. Abstraction d. Inheritance
- 6. What is the behaviour aspect of an object represented by?
  - a. Member Functions b. Data Members
  - c. Both a and b d. None of these
- 7. What is the ability of an object to take on many forms called?
  - a. Polymorphism b. Encapsulation
  - c. Abstraction d. Inheritance
- 8. What is the term that is used to represent hierarchical relationship of generalization?
  - a. Polymorphism b. Encapsulation
  - c. Abstraction d. Inheritance

Question Answer

- Name the art of implementing Encapsulation in Object Oriented Programming.
  - a. Polymorphism
    b. Encapsulation
  - c. Abstraction
    d. class
- What is meant by state of an object?
  - a. Functions of the object
    b. Data Members of the object
  - c. Content of an object
    d. All of these